

Strive Academy Adaptive Learning Protocol

Why an Integrated Approach?

Conventional school curricula often segment subjects (history, reading, vocabulary, science) into isolated time slots. Cognitive research shows that the brain builds associative memory networks rather than separate “filing cabinets,” so learning sticks better when information is connected and meaningful. 4MAT creator Bernice McCarthy emphasizes that the **4MAT System** honors the unique learning style each student brings while encouraging students to master the entire cycle of learning styles. This means that successful instruction helps students stretch beyond their preferred style and experience the full learning cycle.

Stacking functions—a concept borrowed from permaculture—means designing lessons that accomplish multiple goals simultaneously. When students explore a theme like Ancient Egypt, reading passages, vocabulary, geography, math and art all reinforce each other, strengthening literacy, comprehension, cultural understanding and retention. Instead of adding more content, we weave existing subjects together to create immersive units.

Foundations

4MAT Learning Cycle

The 4MAT model organizes instruction into a four-phase cycle:

1. **WHY (Emotional relevance)** – connect new content to students’ lives; answer “why does this matter?”
2. **WHAT (Core knowledge)** – present facts, principles and concepts.
3. **HOW (Application)** – provide hands-on practice, procedures and processes.
4. **WHAT IF (Innovation & transfer)** – allow students to create, experiment and transfer learning to new contexts.

McCarthy notes that although students naturally prefer different styles, they also need experience in the other styles to complete a natural learning cycle. Balancing these phases ensures that lessons move from motivation and understanding to practice and creativity. See the diagram below for a visual representation of the cycle.

Retrieval Practice and Spaced Learning

Active retrieval strengthens memory far more than passive re-reading. A 2025 study in **Frontiers in Psychology** showed that in primary-school settings, a testing procedure that required students to recall information until they achieved 100 % accuracy produced better learning than simply re-reading history texts. The same article explains that retrieval practice deepens comprehension and consolidates information in long-term memory, whereas re-reading can create an illusion of competence. Moreover, combining retrieval practice with distributed (spaced) practice can further enhance retention.

Spaced practice works by revisiting material across longer intervals rather than cramming. Research summarizing the **spacing effect** notes that spacing learning material across various temporal intervals builds stronger memory, alleviating the cognitive problems caused by cramming. Meta-analyses show that spaced repetition improves recall efficiency and long-term retention.

Lean and Kaizen (Continuous Improvement)

Lean thinking focuses on maximizing value and eliminating waste. The Lean Enterprise Institute notes that continuous improvement (often called *kaizen*) is a philosophy of making small, incremental changes to processes to remove waste and increase efficiency and quality. Originating from Toyota's manufacturing system, lean emphasizes smooth "flow" (work moves efficiently from one step to the next) and a "pull" system (work is performed when needed). Equally important is empowering everyone—front-line workers or students—to identify and solve problems.

Kaizen's continuous improvement mindset fits well with education: students regularly reflect on what was hard, what strategies worked and how they can improve, turning learning into an iterative process. Instead of finalizing a curriculum and moving on, teachers and students collaboratively refine learning routines and materials.

EOS and Organizational Structure

The **Entrepreneurial Operating System® (EOS)** provides a framework for leadership teams to clarify their vision, gain traction on priorities and become cohesive. EOS introduces tools such as accountability charts, scorecards and regular "Level 10" meetings for structured reflection and alignment. For Strive Academy, EOS principles help ensure that teachers and administrators share a clear vision, have defined roles and regularly monitor progress.

The STRIVE™ Protocol (Remade)

The STRIVE™ protocol combines stacking functions, 4MAT, retrieval practice, spaced learning, lean/kaizen and EOS principles into a coherent method for designing learning experiences. Each letter encapsulates a key element:

Letter	Meaning	How it connects
S – Stack Functions	Design lessons that accomplish multiple goals simultaneously. For example, reading assignments incorporate vocabulary, geography and writing about a common theme. This reduces fragmentation and increases meaningful connections.	Aligns with 4MAT's emphasis on <i>WHY</i> by making content relevant, and with lean by eliminating redundant lessons.
T – Teach through 4MAT	Structure each lesson using the 4MAT cycle: start with an engaging <i>WHY</i> hook, deliver <i>WHAT</i> core knowledge, provide <i>HOW</i> practice, invite <i>WHAT IF</i> creative transfer, and check for <i>MASTERY</i> .	Ensures that different learning styles are addressed and students complete a natural learning cycle.
R – Retrieval + Spaced Practice	Use frequent retrieval (quizzes, flashcards, presentations, teaching others) and spaced repetition to strengthen memory. Schedule short review sessions over days or weeks rather than one large study session.	Aligns with research showing that retrieval and spacing improve long-term retention and comprehension.
I – Integrated Learning Paths	Organize curriculum around themes (e.g., “Civilisations,” “Weather,” “Entrepreneurship”) that cut across subjects. Each subject uses the same vocabulary and context but explores it through different lenses.	Encourages associative memory networks; supports cross-disciplinary projects; reduces cognitive load.
V – Variable Modes	Provide multiple ways to engage: audio narration, visuals, kinesthetic activities, discussion, simplified summaries and projects. Offer accommodations and extensions for neurodivergent learners (ADHD, autism, dyslexia).	Addresses diverse learning preferences, supports universal design for learning and fosters equitable access.
E – Evaluate & Evolve (Kaizen)	Regularly evaluate mastery through retrieval, projects and teaching back. Encourage students to reflect on what was challenging and how to improve. Teachers use feedback loops to refine lessons and remove unnecessary work, embodying continuous improvement.	Aligns with Lean/Kaizen; ensures the curriculum adapts and remains effective.

Designing a Strive Academy Unit

1. **Choose a Theme and Goals:** Select a theme connected to standards and student interests (e.g., “The Roman Empire,” “Sustainable Energy,” or “Local Ecosystems”). Clearly define learning outcomes across subjects.
2. **Create a 4MAT Lesson Cycle:**
 - *WHY* hook: Use a story, problem, or real-life connection to pique curiosity.
 - *WHAT* knowledge: Present key facts, vocabulary and concepts (through readings, videos, guest talks).
 - *HOW* practice: Provide hands-on activities (experiments, models, simulations, role-plays).
 - *WHAT IF* transfer: Challenge students to innovate or apply knowledge (design a water system like Roman aqueducts; debate a modern ethical dilemma).
3. **Stack Functions:** Integrate reading, vocabulary, geography, math, science and art into the same theme. For example, while studying “Volcanoes,” students might:
 - Read narrative accounts of eruptions (reading fluency).
 - Learn and use vocabulary like “magma,” “lava,” and “plate tectonics” (vocabulary).
 - Map major volcanic regions (geography).
 - Calculate eruption volumes or slopes (math).
 - Model eruptions with baking soda and vinegar (science).
 - Create posters or digital presentations (art/communication).
4. **Plan Retrieval & Spacing:** Build a schedule of short retrieval activities:
 - Daily or every-other-day timed readings of the text.
 - Flashcards or quizzes reviewing vocabulary.
 - Quick write-ups or oral summaries.
 - Cumulative quizzes that revisit older material.Vary the spacing intervals to encourage long-term retention.
5. **Provide Variable Modes:** Offer multiple modalities:
 - Audio versions of readings.
 - Visual diagrams and maps.
 - Movement-based activities (e.g., acting out plate movements).
 - Simplified summaries for students needing extra support.
 - Extended challenges for advanced learners.
6. **Evaluate & Reflect:**
 - Use retrieval-based assessments (short-answer, teaching-back, projects) rather than only multiple-choice.
 - Have students journal about what strategies helped them.
 - Conduct quick “kaizen” meetings where the class discusses what could improve the lesson.
 - Teachers adjust the plan accordingly, eliminating busy work and adding clarity.
7. **Document & Share:**
 - Use an accountability chart (EOS concept) to assign roles: teacher, student leader, group roles.
 - Maintain a scorecard to track learning goals, retrieval scores and improvements.

- Hold regular check-ins (“Level 10” meetings adapted for education) to align on vision and progress.

Simplifying Complex Concepts for Kids

To make these ideas accessible:

- Use kid-friendly analogies: compare the 4MAT cycle to a video game level progression—each level unlocks new abilities.
- Present lean/kaizen as “tidying your backpack regularly so nothing gets lost and you can find things quickly.”
- Illustrate retrieval practice as “playing memory games to strengthen your brain muscles.”
- Incorporate stories and characters to personify abstract ideas (e.g., “Kaizen the Cat always asks ‘How can I make this better?’”).

Toward an AI-Powered Lesson Generator

Strive Academy can automate lesson creation by developing an AI prompt protocol. Inputs include:

- Grade level and reading level.
- Subject(s) and theme.
- Desired vocabulary.
- Neurodivergent supports (e.g., simplified text, audio).
- Desired outputs (projects, quizzes, discussions).

The AI would output a 4MAT lesson sequence, retrieval and spacing schedule, integrated cross-subject tasks, and evaluation rubrics. Teachers can then review, adjust and implement lessons, ensuring they align with standards and class needs.

Conclusion

The remade STRIVE protocol is a human-centered learning framework that integrates 4MAT’s natural learning cycle, evidence-based retrieval and spaced practice strategies, lean/kaizen continuous improvement and EOS organizational clarity. By stacking functions, connecting subjects, providing varied modes and embedding reflection, Strive Academy can cultivate adaptive learners who understand *why* things matter, know *what* they are learning, can *do* it, and imagine *what if* possibilities. Continuous improvement ensures the curriculum remains dynamic, efficient and responsive to student feedback.